

Adaptive-RAG for Biomedical Question Answering: Balancing Latency and Accuracy via Multi-Modal Routing

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Abstract

Large Language Models (LLMs) show immense promise in biomedical question answering, yet hallucination in high-stakes clinical domains necessitates computationally expensive Retrieval-Augmented Generation (RAG) pipelines. To optimize the trade-off between factual accuracy and operational efficiency, we present an Adaptive-RAG framework that dynamically routes user queries between a parametric LLM and a vector knowledge base. We analyze the evolution of this routing architecture from a baseline k -Nearest Neighbors heuristic which fused uncertainty metrics (entropy) with semantic embeddings to advanced machine learning classifiers, including a Llama-3 LoRA adapter, DistilBERT, and XGBoost. Results show our micro-model routers exhibit emergent, dynamically tunable cost-optimization capabilities. By calibrating the risk-tolerance threshold, our XGBoost framework bypasses heavy retrieval for 24% of queries, saving token compute and reducing latency* while maintaining a competitive accuracy.

1 Introduction

As Large Language Models (LLMs) are increasingly integrated into critical fields such as healthcare and biomedical research, mitigating unfaithful or hallucinated outputs has become paramount. Standard Retrieval-Augmented Generation (RAG) effectively addresses the knowledge-cutoff problem by grounding the LLM’s generation in verified, external databases. By retrieving and injecting relevant medical abstracts directly into the context window, RAG provides a robust safety net against hallucinations.

However, standard RAG applies a "one-size-fits-all" methodology, executing heavy vector database retrievals and drastically expanding the input context even for rudimentary queries that the LLM could easily answer from its own internal weights.

This introduces latency bottlenecks and excessive token consumption, making enterprise-scale deployment financially and computationally expensive. While adaptive routing frameworks exist, many rely on computationally heavy "LLM-as-a-judge" mechanics.

Furthermore, our evaluations revealed that attempting to route queries using simple geometric algorithms (such as k -Nearest Neighbors operating on fused semantic and entropy vectors) catastrophically fails due to the curse of dimensionality.

To solve this, we propose a dynamically tunable Adaptive-RAG pipeline tailored specifically for the biomedical domain. Our method logic rests on completely decoupling the "routing brain" from the "generative brain." Rather than relying on rigid heuristics or massive models, we evaluate a spectrum of parameter-efficient machine learning routers including a LoRA adapter, a DistilBERT micro-transformer, and a microsecond-latency XGBoost classifier trained on dense sentence embeddings. These routers analyze the hallucination risk of an incoming question prior to generation, acting as a triage layer that dictates whether to trigger a document retrieval or to trust the parametric LLM knowledge.

Through a comparative study of three adaptive routers against standard parametric and RAG baselines, we map the Pareto frontier of risk versus compute optimization. Our analysis highlights the operational trade-offs of each model. Ultimately, we establish that by calibrating the confidence threshold of the fast XGBoost router, the system successfully bypasses the vector database 24% of the time. This framework achieves the targeted midpoint between an "All-RAG" and "All-LLM" approach, saving computational tokens without compromising diagnostic accuracy.

2 Related Work

Retrieval-Augmented Generation and Active Retrieval: RAG has emerged as the standard methodology for mitigating hallucinations by grounding Large Language Models (LLMs) like Llama-3 (11) in external knowledge. Advanced iterations of this concept, such as Forward-Looking Active Retrieval (FLARE) (5), iteratively trigger retrieval during the generation process whenever the model predicts upcoming tokens with low confidence. While FLARE effectively leverages parametric and non-parametric knowledge, its token-by-token evaluation during generation incurs massive latency overhead. In contrast, our work decouples the routing decision from generation entirely. By evaluating query complexity *before* the LLM initiates a forward pass, we prevent the computational prefill cost that FLARE inevitably absorbs.

Similarly, Jeong et al. (4) introduced the Adaptive-RAG framework, utilizing a smaller Language Model to classify query complexity and dynamically route queries between no-retrieval, single-step, or multi-step retrieval pathways. While our architecture shares this foundational routing philosophy, it differs in domain application and computational granularity. We specifically target the biomedical domain (6; 9). Furthermore, rather than relying exclusively on an LM classifier, we also introduce low-latency classical models (XGBoost operating on Sentence-BERT (10) embeddings).

Hallucination Detection Paradigms: Traditional approaches to hallucination detection rely on post-hoc analysis. Frameworks evaluating lexical overlap or utilizing classifiers trained on beyond-ROUGE subspace features (7) assess the output against datasets like HaluEval (8). However, post-hoc methods treat the LLM as a "Black-Box" and cannot preemptively stop hallucinations. To address this, the INSIDE framework (1) demonstrated that an LLM's internal representations (e.g., logits, token entropies) inherently retain the power to detect hallucinations. Our Phase 1 baselines empirically validated this "White-Box" approach; however, extracting internal states requires the LLM to process the query, negating any RAG latency savings. Our proposed machine-learning routers bridge this gap, achieving predictive hallucination detection without requiring a generative forward pass.

Parameter-Efficient Adaptation: Fine-tuning

an entire LLM to act as a router is computationally prohibitive. Techniques like Low-Rank Adaptation (LoRA) (3) freeze the pre-trained model weights and inject trainable rank decomposition matrices, vastly reducing the parameter overhead. We integrate LoRA as one of our comparative routing baselines against standard dense vector retrieval via FAISS (2). While LoRA demonstrates strong classification capabilities, our findings ultimately contrast PEFT architectures with microsecond-latency decision trees, proving that for binary RAG routing, classical ML paired with dense semantic embeddings outperforms parameter adapters in operational efficiency.

3 Problem Description

Large Language Models (LLMs) lack the inherent factual reliability required for autonomous biomedical question answering. While standard Retrieval-Augmented Generation (RAG) mitigates clinical risk by grounding responses in external literature, executing vector database searches for every query introduces high latency and token costs at an enterprise scale.

In Phase 1 of this project, we evaluated multiple hallucination detection strategies. We demonstrated that "White-Box" internal probing (analyzing the LLM's hidden states and token entropies) vastly outperformed classical machine learning applied to surface-level text. However, utilizing internal states presents a critical operational paradox: to confidently detect a hallucination, the LLM must first execute a computationally expensive generative forward pass, entirely defeating the latency goals of an adaptive system.

Therefore, Phase 2 addresses the need for a preemptive, low-latency routing mechanism capable of triaging queries *before* generation begins. Formally, for an incoming query Q , the system must choose between a fast Parametric pathway (cost C_p , latency t_p) and a highly accurate RAG pathway (cost $C_r \gg C_p$, latency $t_r \gg t_p$). Our objective is to design a binary routing function $R(Q) \in \{0, 1\}$ that maximizes factual accuracy while maximizing the RAG bypass rate, eliminating the need for a generative forward pass on safe queries.

4 Methods

Our methodology leverages multiple specialized datasets and models to construct and evaluate the Adaptive-RAG pipeline. We utilize the MedHallu

dataset (9)—a comprehensive benchmark for detecting medical hallucinations—to train our predictive routing models. For end-to-end system evaluation, we use the expert-annotated pqa_labeled split of the PubMedQA corpus (6). All generation is powered by the Llama-3.1-8B-Instruct model (11). Below, we detail the pipeline across three distinct developmental phases.

4.1 Phase 1: Hallucination Detection via Internal Probing

In this initial phase, we established the baseline capability of Llama-3.1 to self-report uncertainty by extracting its hidden mechanistic states, serving as our "White-Box" oracle.

4.1.1 Data Processing and Extraction

We sampled 1,000 clinically validated questions from the PubMedQA dataset. For each query q and its associated medical abstract c , we prompted Llama-3.1 to generate a draft response using a standardized Chain-of-Thought template: Context: {c} \n Question: {q} \n Let's think step by step to determine if the medical answer is Yes, No, or Maybe:. During the forward pass, we intercepted the model's generation logits. For each generated token t_i , we applied a softmax function over the vocabulary distribution to extract token-level probabilities, computing the Shannon Entropy: $H(t_i) = -\sum p(x_j) \log p(x_j)$.

From the sequence of token entropies, we computed two scalar values for each query: **Mean Entropy** (μ_H) to represent overall generative uncertainty, and **Varentropy** (σ_H^2) to capture localized spikes indicating potential fabricated entities. Ground truth labels were assigned by verifying if the conclusion of the model's generated text matched the expert-annotated final_decision (0 = Factual/Safe, 1 = Hallucination/Trigger RAG).

4.2 Phase 2: Knowledge Base Construction and Multi-Modal Routing

4.2.1 FAISS Corpus Preparation

We decoupled our evaluation dataset from our knowledge base by embedding an external corpus of over 211,000 published medical abstracts. Using the all-MiniLM-L6-v2 Sentence-BERT model (10), we converted each abstract into a 384-dimensional dense vector. These vectors were indexed using Facebook AI Similarity Search (FAISS) (2) with an IndexFlatL2 configuration,

allowing for rapid k -Nearest Neighbor (k -NN) retrieval based on Euclidean distance.

4.2.2 The k -NN State Vector Router

We constructed a 386-dimensional "State Vector" for each query in our 1,000-sample Phase 1 history by concatenating its 384D semantic embedding with its extracted μ_H and σ_H^2 scalars. To prevent geometric overshadowing by the high-dimensional text space, the two entropy scalars were artificially scaled by a weight factor of 50.0. During execution, an incoming query was embedded, and the router queried the FAISS index ($k = 5$). If any of the retrieved historical neighbors possessed a hallucination label of 1, the system triggered the RAG pathway. (Note: As discussed in the Problem Description, this linear approach ultimately failed, prompting Phase 3).

4.3 Phase 3: Adaptive Machine Learning Routers

To resolve the dimensionality and latency bottlenecks, we developed three parametric machine learning routers trained to predict the binary routing decision (0 = Parametric, 1 = RAG) without requiring an LLM forward pass.

4.3.1 System A: Llama-3 LoRA Adapter

We applied Parameter-Efficient Fine-Tuning (PEFT) (3) to the base Llama-3.1-8B model. We attached a sequence classification head (num_labels=2) and fine-tuned it on the MedHallu training split. Inputs were tokenized and truncated to a maximum length of 2048 tokens. During inference, we extracted the softmax probabilities from the sequence classifier logits. If the probability of class 1 exceeded 0.70, the RAG pathway was activated.

4.3.2 System B: DistilBERT Micro-Transformer

For a lightweight neural alternative, we fine-tuned distilbert-base-uncased (66M parameters). The training methodology mirrored the LoRA setup but with a strictly constrained max_length of 512 tokens to optimize for millisecond inference. Softmax logits were extracted, and a threshold of 0.70 was applied to determine the routing path.

4.3.3 System C: Classical XGBoost Classifier

For ultra-low latency routing, we abandoned neural sequence classification in favor of classical decision trees. Incoming queries from the MedHallu

dataset were embedded into 384-dimensional vectors using Sentence-BERT. An XGBoost classifier was trained on these dense vectors using a binary logistic learning objective. During inference, we utilized the `predict_proba()` function to extract the raw hallucination risk score.

4.4 System Benchmark Execution

The final shootout evaluated all architectures over a 100-query random sample from PubMedQA. For Standard RAG, queries were embedded, the top 3 abstracts were retrieved from FAISS, and the context was injected into the Llama-3 prompt. Token counts were tracked by measuring the `input_ids` tensor shape for both the parametric and context-augmented prompts, while execution latency was recorded via the Python time module to evaluate operational efficiency.

5 Experimental Results

5.1 Phase 1: Hallucination Detection Baselines

Models and Baselines. To establish a hallucination detection baseline, we evaluated surface-level lexical metrics (N-gram overlap) and classical machine learning classifiers (SVM and XGBoost trained on beyond-ROUGE features (7)) against "White-Box" Internal Probing methodology on the base Llama-3.1-8B-Instruct model.

Evaluation Sets and Metrics. Evaluation was conducted across standard QA, Dialogue, and Summarization splits of the HaluEval (8) dataset. Performance was measured utilizing Area Under the Receiver Operating Characteristic curve (AUROC) and Macro-F1 scores.

Results and Discussion. As demonstrated in Table 1, while XGBoost performed adequately on standard QA (AUROC 0.84), traditional post-hoc methods suffered severe degradation on complex generative tasks. Conversely, Internal Probing demonstrated superiority (AUROC of 0.95 on QA). This validated our hypothesis that internal states encode uncertainty; however, extracting them requires a forward pass, leading to the Phase 2 RAG architecture.

5.2 Phase 2: RAG Backend and k -NN Router

Dataset and Implementation. We embedded 211,000 medical abstracts into a FAISS index using `all-MiniLM-L6-v2`. For routing, we evaluated

Methodology	QA		Dialogue		Summary	
	AUC	F1	AUC	F1	AUC	F1
N-Gram	0.79	0.85	0.60	0.63	0.50	0.49
SVM	0.72	0.78	0.63	0.69	0.46	0.55
XGBoost	0.84	0.90	0.66	0.71	0.48	0.46
Internal Probe	0.95	0.98	0.82	0.88	0.93	0.96

Table 1: Phase 1 Baseline Hallucination Detection Metrics.

a k -NN ($k = 5$) algorithm acting upon a 386-dimensional State Vector (384D semantic text + 2D scaled entropy scalars).

Results and Discussion. The k -NN geometric router failed. Out of 100 benchmarked queries, the system routed 90% to the Parametric (Vanilla LLM) pathway, entirely bypassing the RAG database and yielding an accuracy of 62.6%. The L_2 Euclidean distance caused the 384 text dimensions to completely mathematically overshadow the 2 entropy dimensions, forcing the system to cluster queries by medical topic rather than hallucination risk.

Routing Path	Traffic Split	Path Acc.
Parametric (LLM)	90.0% (90)	64.0%
RAG (Database)	10.0% (10)	50.0%
Overall System	-	62.6%

Table 2: benchmarking results demonstrating the k -NN performance.

5.3 Phase 3: Adaptive ML Routing Shootout

Models and Training Setup. We evaluated three architectures to replace the failed k -NN approach:

- **System A (Llama-3 LoRA):** PEFT adaptation of Llama-3.1-8B, rank $r = 16$, $\alpha = 32$.
- **System B (DistilBERT):** 66M parameters, fine-tuned sequence classification head.
- **System C (XGBoost):** Trained on 384D Sentence-BERT embeddings.

All routers were trained on the MedHallu (9) dataset to predict binary hallucination risk.

Benchmark Protocol and Metrics. We executed a 100-query benchmark using the `pqa_labeled` PubMedQA dataset. We measured end-to-end System Accuracy, Execution Latency (seconds), Total Tokens Processed, and the RAG Bypass Percentage.

Architecture	Accuracy	Latency* (s)	Tokens Read	RAG Bypassed
1. Vanilla LLM	53%	194.92	2,758	100%
2. Standard RAG	78%	200.10	33,722	0%
5. System A (Llama-3 LoRA)	61%	202.54	15,946	58%
4. System B (DistilBERT)	51%	195.98	5,247	92%
3. System C (XGBoost)	69%	199.79	26,003	24%

* Latency was measured in a localized, single-node environment. In production architectures, vector databases are typically hosted remotely, meaning standard RAG would incur substantial network I/O overhead, further amplifying the latency savings of the XGBoost bypass.

Table 3: Final Benchmark Results across 100 PubMedQA Queries.

Results and Discussion. Table 3 illustrates the operational trade-offs. Standard RAG establishes the accuracy ceiling (78%) but consumes an immense 33,722 tokens with high latency. DistilBERT (System B) bypassed the DB 92% of the time, dropping accuracy to 51%. The XGBoost model (System C) proved superior, successfully bypassing the heavy retrieval for 24% of queries, saving over 7,700 compute tokens while maintaining a highly competitive 69% accuracy.

6 Conclusion and Future Work

This project successfully demonstrates that "Always-On" Retrieval-Augmented Generation is not a strict necessity for safe biomedical question answering. By decoupling the routing decision from the generative process, we established that lightweight machine learning classifiers can effectively preempt LLM hallucinations. While our initial geometric k -NN router failed due to high-dimensional overshadowing, transitioning to parametric models proved highly effective. Specifically, XGBoost classifier operating on dense semantic embeddings bypassed the retrieval database 24% of the time, drastically reducing token consumption and API latency while preserving diagnostic accuracy.

Future work should explore a "Cascade Routing" protocol. In such an architecture, an XGBoost model would serve as the primary microsecond gatekeeper for obvious edge cases. Borderline or highly uncertain queries would then be escalated to a moderately sized micro-transformer (e.g., DistilBERT) for a deeper semantic check before ultimately triggering the FAISS backend. This multi-tiered approach would further optimize the operational efficiency and safety of autonomous medical AI.

Individual Contributions

Faisal: Engineered the FAISS retrieval backend, embedded 211,000 abstracts, and built the automated benchmarking shootout loop. Developed the XGBoost and DistilBERT micro-routing scripts to optimize system latency.

Abin: implemented the Llama-3 LoRA adapter (System A), including the PEFT configuration, sequence classification head attachment, and inference threshold calibration and also contributed to the related work section, particularly on parameter-efficient fine-tuning and active retrieval methods.

Priyal: Implemented a curriculum-based fine-tuning strategy for the Llama-3 LoRA adapter, optimizing the rank-decomposition matrices ($r = 16, \alpha = 32$) on specialized medical hallucination benchmarks to achieve precise query-complexity classification.

Anton: Led the end-to-end performance validation of the Adaptive-RAG pipeline, establishing the ground-truth hallucination metrics used to train the Llama-3 LoRA and XGBoost routers and benchmarking their effectiveness against high-stakes biomedical datasets.

Garima: Conducted sensitivity analysis and synthetic stress tests on the Adaptive-RAG pipeline, utilizing adversarially deceptive clinical scenarios to map the Pareto frontier of risk versus compute optimization.

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